Program	Computer Science and Engineering	**Semester**	3
Course Name	Data Structures with Python	**Test**	IV
Course Code	25CS33I	**Duration**	180 min
Name of the Course Coordinator:	Vinay Kumar	**Marks**	50

Questions	CO	Marks
Problem Statement: Library Management using Stack, Queue, Linked Lists and Binary Tree Write a menu-driven program, using user-defined classes (UDT) to implement the ADTs, that performs the following tasks: 1. **Stack: Recently Returned Books (LIFO)** o Implement a Stack class with push(), pop(), peek(), is_empty() and display(). o Push each returned book, remove the last returned book and view the top book. o Display a message when the stack is empty. 2. **Queue: Book Request Counter (FIFO)** o Implement a Queue class with enqueue(), dequeue(), is_empty() and display(). o Add a student to the request queue, serve the student at the front and display the waiting students. o Display a message when no student is waiting. 3. **Singly Linked List: Book Catalogue** o Create a Node class and a Singly Linked List class storing Book ID and Title. o Insertion at the beginning, at the end and in the middle. o Deletion from the beginning, from the end and from the middle. o Traversal and searching for a book by Book ID. 4. **Doubly Linked List: Member Borrowing History** o Create a Node class with previous and next pointers and a Doubly Linked List class. o Insertion at the beginning, at the end and in the middle; deletion from the beginning, from the end and from the middle. o Traversal in the forward and in the backward direction. 5. **Binary Tree: Book ID Index** o Create a binary tree using a Node class by inserting the Book IDs 50, 30, 70, 20, 40, 60, 80. o Search for a Book ID and delete a node entered by the user. o Display In-order, Pre-order and Post-order traversals using recursion. o Display the same traversals using an iterative method (use a stack) and compare with the recursive output. 6. **Output and Error Handling** o Display the result of every operation performed from the main menu. o Handle errors such as empty stack/queue, invalid position, non-existent Book ID and invalid input.	**1,3,4,5**	**50**

Scheme of assessment (customize scheme as per requirement of your problem statement)

a) Program Design and Conceptual Clarity (Clear identification of the key concepts, Explanation of the logic and methodology used and organization of the program.) - 10

b) Implementation and Execution - 30

c) Best Practices (Code Readability and Error Handling) - 10

Total Marks 50

Signature of the Course Coordinator

Signature of the HOD